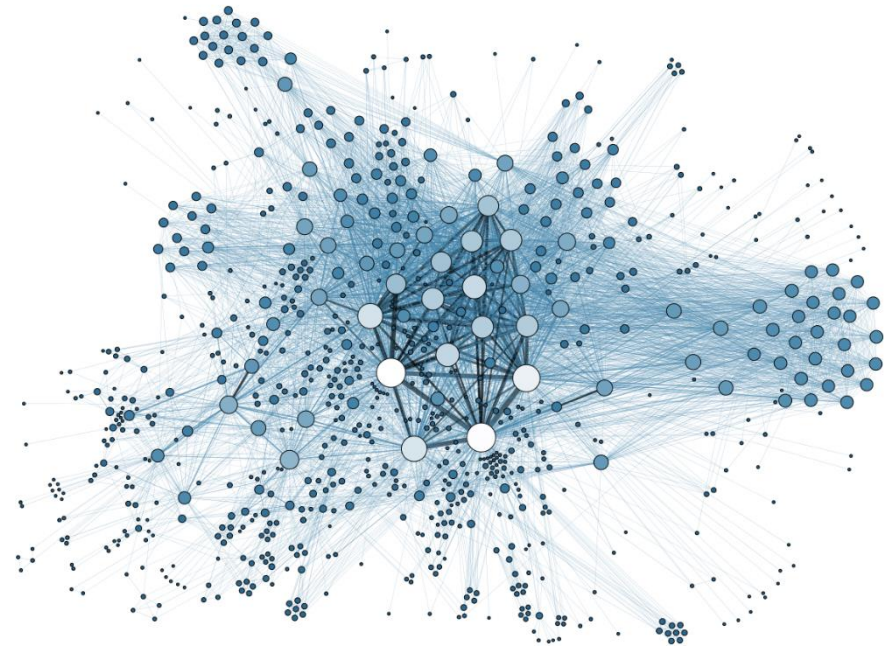




Network Measures

SE4095-Social Network: Theory and Practice



Why Do We Need Measures?

- Who are the central figures (influential individuals) in the network?
 - **Centrality**
- What interaction patterns are common in friends?
 - **Reciprocity and Transitivity**
 - **Balance and Status**
- Who are the like-minded users and how can we find these similar individuals?
 - **Similarity**
- To answer these and similar questions, one first needs to define measures for quantifying **centrality, level of interactions, and similarity**, among others.

Centrality

Centrality defines how important a node is within a network

**Centrality in terms of those who you
are connected to**

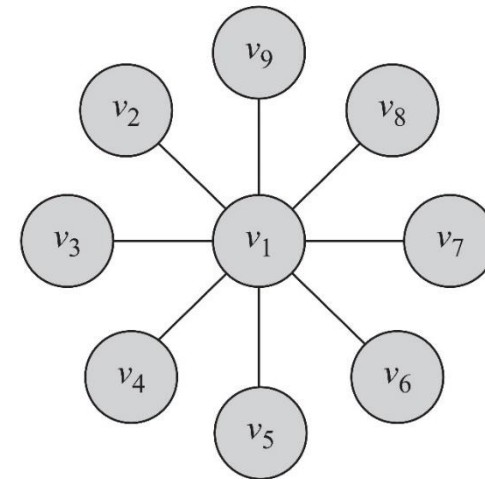
Degree Centrality

- **Degree centrality:** ranks nodes with more connections higher in terms of centrality

$$C_d(v_i) = d_i$$

- d_i is the degree (number of friends) for node v_i
 - i.e., the number of length-1 paths (can be generalized)

In this graph, degree centrality for node v_1 is $d_1=8$ and for all others is $d_j = 1, j \neq 1$



Degree Centrality in Directed Graphs

- In directed graphs, we can either use the in-degree, the out-degree, or the combination as the degree centrality value:
- In practice, mostly in-degree is used.

$$C_d(v_i) = d_i^{\text{in}} \quad (\textit{prestige})$$

$$C_d(v_i) = d_i^{\text{out}} \quad (\textit{gregariousness})$$

$$C_d(v_i) = d_i^{\text{in}} + d_i^{\text{out}}$$

d_i^{out} is the number of outgoing links for node v_i

Normalized Degree Centrality

- Normalized by the maximum possible degree

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

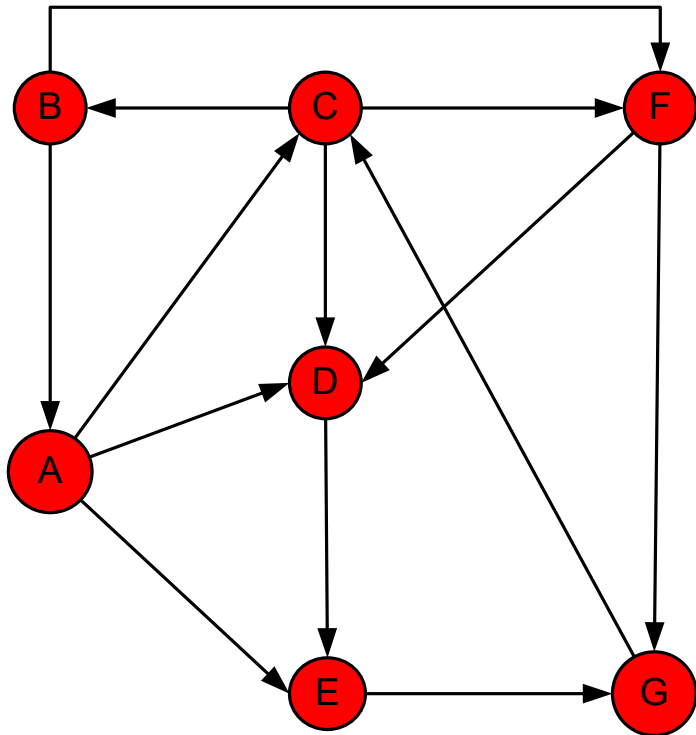
- Normalized by the maximum degree

$$C_d^{\text{max}}(v_i) = \frac{d_i}{\max_j d_j}$$

- Normalized by the degree sum

$$C_d^{\text{sum}}(v_i) = \frac{d_i}{\sum_j d_j} = \frac{d_i}{2|E|} = \frac{d_i}{2m}$$

Degree Centrality (Directed Graph)Example



Node	In-Degree	Out-Degree	Centrality	Rank
A	1	3	1/2	1
B	1	2	1/3	3
C	2	3	1/2	1
D	3	1	1/6	5
E	2	1	1/6	5
F	2	2	1/3	3
G	2	1	1/6	5

Normalized by the maximum possible degree

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

What we want to know

How many people does this person connect/follow?

How many people connect/follow this person?

Overall connectivity in an undirected network

Who receives information from many people?

Who can send information to many people?

Use

Out-degree

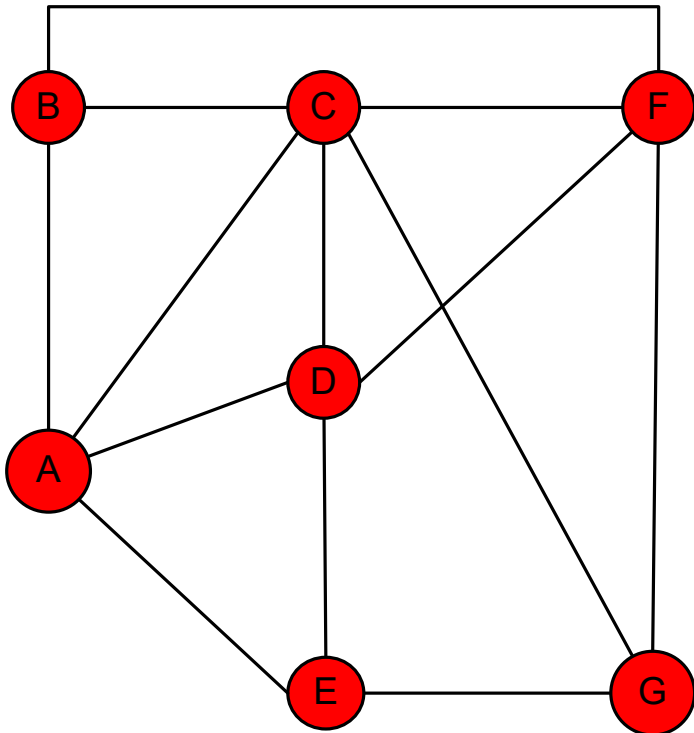
In-degree

Degree

In-degree

Out-degree

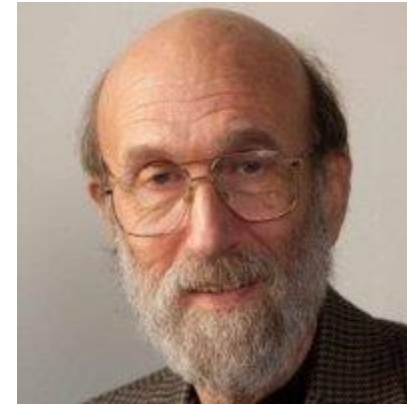
Degree Centrality (undirected Graph) Example



Node	Degree	Centrality	Rank
A	4	$2/3$	2
B	3	$1/2$	5
C	5	$5/6$	1
D	4	$2/3$	2
E	3	$1/2$	5
F	4	$2/3$	2
G	3	$1/2$	5

Eigenvector Centrality

- Having more friends does not by itself guarantee that someone is more important
 - Having more **important friends** provides a stronger signal



Phillip Bonacich

- Eigenvector centrality generalizes degree centrality by incorporating the importance of the neighbors (undirected)
- For directed graphs, we can use incoming or outgoing edges

Formulation

- Let's assume the eigenvector centrality of a node is $c_e(v_i)$ (**unknown**)
- We would like $c_e(v_i)$ to be higher when **important** neighbors (**node v_j with higher $c_e(v_j)$**) point to us
 - Incoming or outgoing neighbors?
 - For incoming neighbors $A_{j,i} = 1$
- We can assume that v_i 's centrality is the summation of its neighbors' centralities
- Is this summation bounded?
 - We have to normalize!

$$c_e(v_i) = \sum_{j=1}^n A_{j,i} c_e(v_j)$$

λ : some fixed constant

$$c_e(v_i) = \frac{1}{\lambda} \sum_{j=1}^n A_{j,i} c_e(v_j)$$

Eigenvector Centrality (Matrix Formulation)

Their centralities are:

$$c_e(A), c_e(B), c_e(C), c_e(D)$$

We put them into a vector:

$$C_e = \begin{bmatrix} c_e(A) \\ c_e(B) \\ c_e(C) \\ c_e(D) \end{bmatrix}$$

The slide writes:

$$C_e = (C_e(v_1), C_e(v_2), \dots, C_e(v_n))^T$$

The T simply means **transpose**, meaning we are treating them as a column vector.

Eigenvector Centrality (Matrix Formulation)

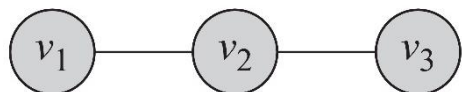
- Let

- Let $\mathbf{C}_e$ be a column vector of size n (where n is the number of nodes in the graph)
- This means that $\mathbf{C}_e$ is an eigenvector of A^T (or A when undirected) and λ is the corresponding eigenvalue
- Which eigenvalue-eigenvector pair should we choose?

$$\rightarrow \lambda \mathbf{C}_e = A^T \mathbf{C}_e$$

- This means that $\mathbf{C}_e$ is an eigenvector of adjacency matrix A^T (or A when undirected) and λ is the corresponding eigenvalue
- Which eigenvalue-eigenvector pair should we choose?

Eigenvector Centrality: Example 1



$$\lambda \mathbf{C}_e = A \mathbf{C}_e \quad (A - \lambda I) \mathbf{C}_e = 0 \quad \mathbf{C}_e = [u_1 \ u_2 \ u_3]^T$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 - \lambda & 1 & 0 \\ 1 & 0 - \lambda & 1 \\ 0 & 1 & 0 - \lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 0 - \lambda & 1 & 0 \\ 1 & 0 - \lambda & 1 \\ 0 & 1 & 0 - \lambda \end{vmatrix} = 0$$

$$(-\lambda)(\lambda^2 - 1) - 1(-\lambda) = 2\lambda - \lambda^3 = \lambda(2 - \lambda^2) = 0$$

Eigenvalues are

$$(-\sqrt{2}, 0, +\sqrt{2})$$

Corresponding eigenvector (assuming $\mathbf{C}_e$ has norm 1)

Largest Eigenvalue

$$\begin{bmatrix} 0 - \sqrt{2} & 1 & 0 \\ 1 & 0 - \sqrt{2} & 1 \\ 0 & 1 & 0 - \sqrt{2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{C}_e = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ \sqrt{2}/2 \\ 1/2 \end{bmatrix}$$

Eigenvector Centrality: Example 1

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad C_e = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$\lambda C_e = A C_e$
 $(A - \lambda I) C_e = 0$

$$\begin{bmatrix} 0-\lambda & 1 & 0 \\ 1 & 0-\lambda & 1 \\ 0 & 1 & 0-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\det(A - \lambda I) = -\lambda(\lambda^2 - 1) - 1(-\lambda)$$

$$= -\lambda^3 + \lambda + \lambda$$

$$= 2\lambda - \lambda^3$$

$$= \lambda(2 - \lambda^2) = 0$$

Case 1: $\lambda = 0$
 Case 2: $2 - \lambda^2 = 0$
 $2 = \lambda^2$
 $\pm\sqrt{2} = \lambda$

eigen value
 $\pm\sqrt{2}$

$$\begin{bmatrix} -\sqrt{2} & 1 & 0 \\ 1 & -\sqrt{2} & 1 \\ 0 & 1 & -\sqrt{2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-\sqrt{2} u_1 + u_2 = 0 \quad \text{--- eq 1}$
 $u_1 + -\sqrt{2} u_2 + u_3 = 0 \quad \text{--- eq 2}$
 $u_2 - \sqrt{2} u_3 = 0$

using eq 1.
 $(\sqrt{2} u_2 = \sqrt{2} u_1)$

using eq 3.
 $u_2 = \sqrt{2} u_3$
 $\frac{u_2}{\sqrt{2}} = u_3$
 $\frac{\sqrt{2} u_1}{\sqrt{2}} = u_1 \Rightarrow u_3 = u_1$

Thus we have.
 $u_2 = \sqrt{2} u_1$
 $u_3 = u_1$

Normalize using ~~etc~~ euclidean norm l_1

$$\|C_e\| = \sqrt{u_1^2 + u_2^2 + u_3^2}$$

$$= \sqrt{u_1^2 + u_2^2 + u_3^2}$$

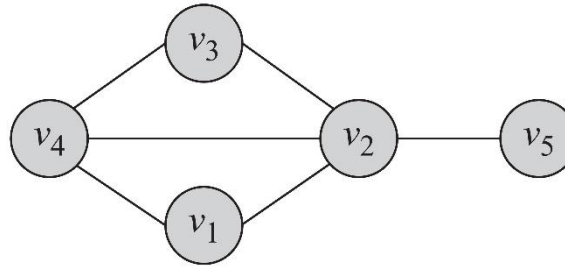
$$= \sqrt{u_1^2 + (\sqrt{2} u_1)^2 + u_1^2}$$

$$= \sqrt{u_1^2 (1+2+1)} \Rightarrow \sqrt{4u_1^2} = 2u_1 \Rightarrow \boxed{u_1 = \frac{1}{2}}$$

$u_2 = \sqrt{2} \frac{1}{2}$
 $\boxed{u_2 = \frac{\sqrt{2}}{2}}$
 $u_3 = u_1$
 $\boxed{u_3 = \frac{1}{2}}$

$$C_e = \begin{bmatrix} 1/2 \\ \sqrt{2}/2 \\ 1/2 \end{bmatrix}$$

Eigenvector Centrality: Example 2



$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \lambda = (2.68, -1.74, -1.27, 0.33, 0.00)$$

↖ Eigenvalues Vector

$$\lambda_{\max} = 2.68 \longrightarrow C_e = \begin{bmatrix} 0.4119 \\ 0.5825 \\ 0.4119 \\ 0.5237 \\ 0.2169 \end{bmatrix}$$

Situation

"Who has the most friends?"

"Who follows the most people?"

"Who has the most followers?"

"Who is connected to other important people?"

"Who is popular because they are connected to popular people?"

Simple measure of direct connectivity

Want importance based on the whole network

Use

Degree Centrality

Out-degree

In-degree

Eigenvector Centrality

Eigenvector Centrality

Degree Centrality

Eigenvector Centrality

Katz Centrality

- A major problem with eigenvector centrality arises when it deals with directed graphs
- Centrality only passes over *outgoing* edges and in special cases such as when a node is in a directed acyclic graph centrality becomes zero
 - The node can have many edge connected to it
- To resolve this problem we add bias term β to the centrality values for all nodes



Elihu Katz

Eigenvector Centrality

$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

Katz Centrality, cont.

$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

Controlling term

Bias term

Rewriting equation in a vector form

$$\mathbf{C}_{\text{Katz}} = \alpha A^T \mathbf{C}_{\text{Katz}} + \beta \mathbf{1}$$

vector of all 1's

Katz centrality: $\mathbf{C}_{\text{Katz}} = \beta (\mathbf{I} - \alpha A^T)^{-1} \cdot \mathbf{1}$

Katz Centrality, cont.

$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

- When $\alpha=0$, the eigenvector centrality is removed and all nodes get the same centrality value β
 - As α gets larger the effect of β is reduced
- For the matrix $(I - \alpha A^T)$ to be invertible, we must have
 - $\det(I - \alpha A^T) \neq 0$
 - By rearranging we get $\det(A^T - \alpha^{-1}I) = 0$
 - This is basically the characteristic equation,
 - The characteristic equation **first** becomes zero when the largest eigenvalue equals α^{-1}

**The largest eigenvalue
is easier to compute
(power method)**

In practice we select $\alpha < 1/\lambda$, where λ is the largest eigenvalue of A^T

- Problem with Katz Centrality:
 - In directed graphs, once a node becomes an authority (high centrality), it passes **all** its centrality along **all** of its out-links
- This is less desirable since not everyone known by a well-known person is well-known
- **Solution?**
 - We can divide the value of passed centrality by the number of outgoing links, i.e., out-degree of that node
 - Each connected neighbor gets a fraction of the source node's centrality

- A university is studying the collaboration network of four students—**A, B, C, and D**—who frequently work together on assignments and research activities. The university wants to identify the student who occupies the **most influential position** in this network.
- At first, the university's coordinator suggests selecting the student who has the **largest number of direct collaborations**. However, another coordinator disagrees. She argues that simply counting collaborations can be misleading because **not all connections are equally valuable**. A student who is connected to highly connected students may have greater influence within the network than someone who has the same number of connections to less-connected students.

Katz Centrality

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- Centrality only passes over *outgoing* edges and in special cases such as when a node is in a directed acyclic graph centrality becomes zero
 - The node can have many edge connected to it
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Katz Centrality, cont.

$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

Controlling term

Bias term

Rewriting equation in a vector form

$$\mathbf{C}_{\text{Katz}} = \alpha A^T \mathbf{C}_{\text{Katz}} + \beta \mathbf{1}$$

vector of all 1's

Katz centrality: $\mathbf{C}_{\text{Katz}} = \beta (\mathbf{I} - \alpha A^T)^{-1} \cdot \mathbf{1}$

Katz Centrality, cont.

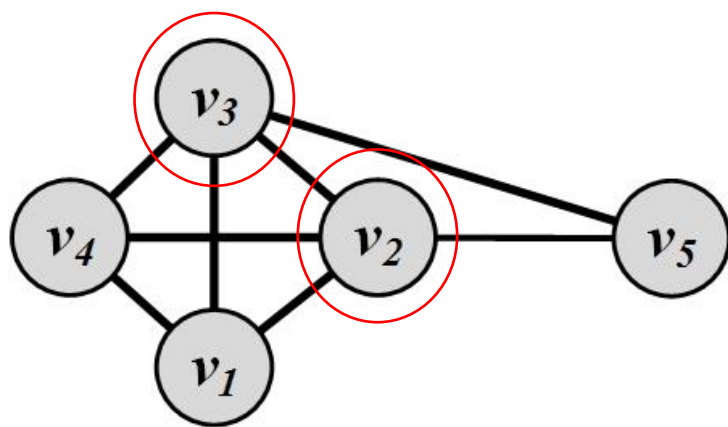
$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

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**The largest eigenvalue
is easier to compute
(power method)**

In practice we select $\alpha < 1/\lambda$, where λ is the largest eigenvalue of A^T

Katz Centrality Example



$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} = A^T$$

- The Eigenvalues are -1.68, -1.0, -1.0, 0.35, 3.32
- We assume $\alpha=0.25 < 1/3.32$ and $\beta = 0.2$

$$\mathbf{C}_{Katz} = \beta(\mathbf{I} - \alpha A^T)^{-1} \cdot \mathbf{1} = \begin{bmatrix} 1.14 \\ 1.31 \\ 1.31 \\ 1.14 \\ 0.85 \end{bmatrix}$$

**Most
important
nodes!**

- Problem with Katz Centrality:
 - In directed graphs, once a node becomes an authority (high centrality), it passes **all** its centrality along **all** of its out-links
- This is less desirable since not everyone known by a well-known person is well-known
- **Solution?**
 - We can divide the value of passed centrality by the number of outgoing links, i.e., out-degree of that node
 - Each connected neighbor gets a fraction of the source node's centrality

PageRank

- In PageRank, a node does not give its entire centrality to every outgoing neighbour. Its centrality is divided among its outgoing links

PageRank, cont.

$$C_p(v_i) = \alpha \sum_{j=1}^n A_{j,i} \frac{C_p(v_j)}{d_j^{\text{out}}} + \beta$$

What if the
degree is
zero?

$$\left\{ \begin{array}{l} d_j^{\text{out}} > 0 \\ D = \text{diag}(d_1^{\text{out}}, d_2^{\text{out}}, \dots, d_n^{\text{out}}) \end{array} \right. \rightarrow \mathbf{C}_p = \alpha A^T D^{-1} \mathbf{C}_p + \beta \mathbf{1}$$

$$\mathbf{C}_p = \beta (\mathbf{I} - \alpha A^T D^{-1})^{-1} \cdot \mathbf{1}$$

Similar to Katz Centrality, in practice, $\alpha < 1/\lambda$, where λ is the largest eigenvalue of $A^T D^{-1}$. In undirected graphs, the largest eigenvalue of $A^T D^{-1}$ is $\lambda = 1$; therefore, $\alpha < 1$.

PageRank, cont.

1. Iterative form:

$$C_p = \alpha A^T D^{-1} C_p + \beta \mathbf{1}$$

2. Rearranging:

$$C_p - \alpha A^T D^{-1} C_p = \beta \mathbf{1}$$

3. Factoring C_p :

$$(I - \alpha A^T D^{-1}) C_p = \beta \mathbf{1}$$

4. Solving for C_p :

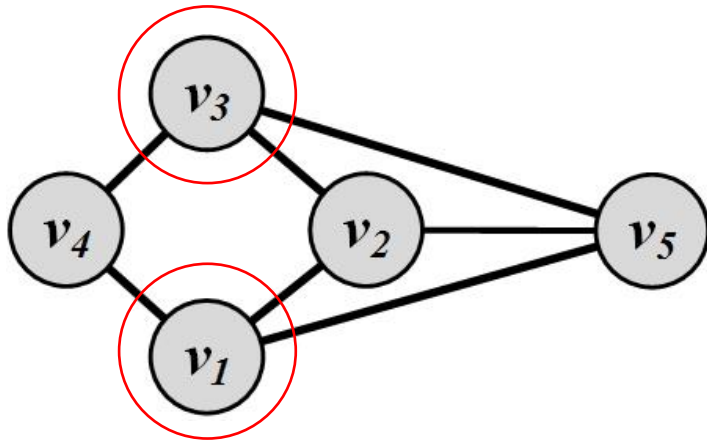
$$C_p = (I - \alpha A^T D^{-1})^{-1} \beta \mathbf{1}$$

5. Final closed-form:

$$C_p = \beta (I - \alpha A^T D^{-1})^{-1} \mathbf{1}$$

PageRank Example

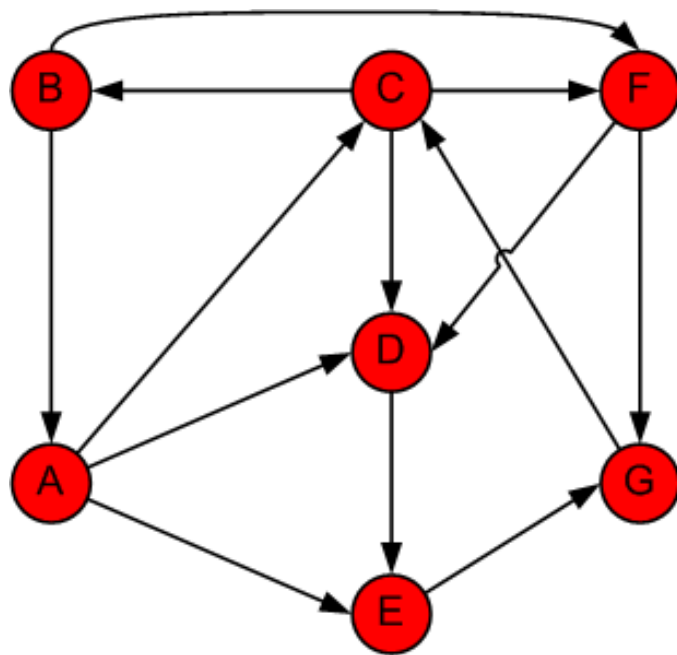
- We assume $\alpha=0.95 < 1$ and $\beta = 0.1$



$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{C}_p = \beta(\mathbf{I} - \alpha A^T D^{-1})^{-1} \cdot \mathbf{1} = \begin{bmatrix} 2.14 \\ 2.13 \\ 2.14 \\ 1.45 \\ 2.13 \end{bmatrix}$$

PageRank Example – Alternative Approach [Markov Chains]



"You don't understand anything until you learn it more than one way"



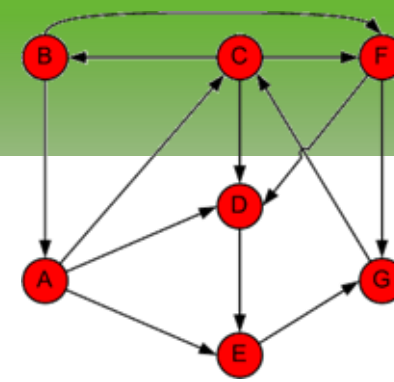
Marvin Minsky (1927-2016)

Using Power Method

$\alpha=1$ and $\beta=0$?

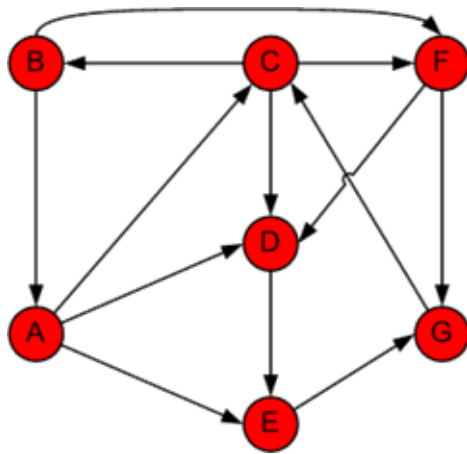
Step	A	B	C	D	E	F	G
0	1/7	1/7	1/7	1/7	1/7	1/7	1/7
1	B/2	C/3	A/3 + G	A/3 + C/3 + F/2	A/3 + D	C/3 + B/2	F/2 + E
	0.071	0.048	0.190	0.167	0.190	0.119	0.214

PageRank: Example



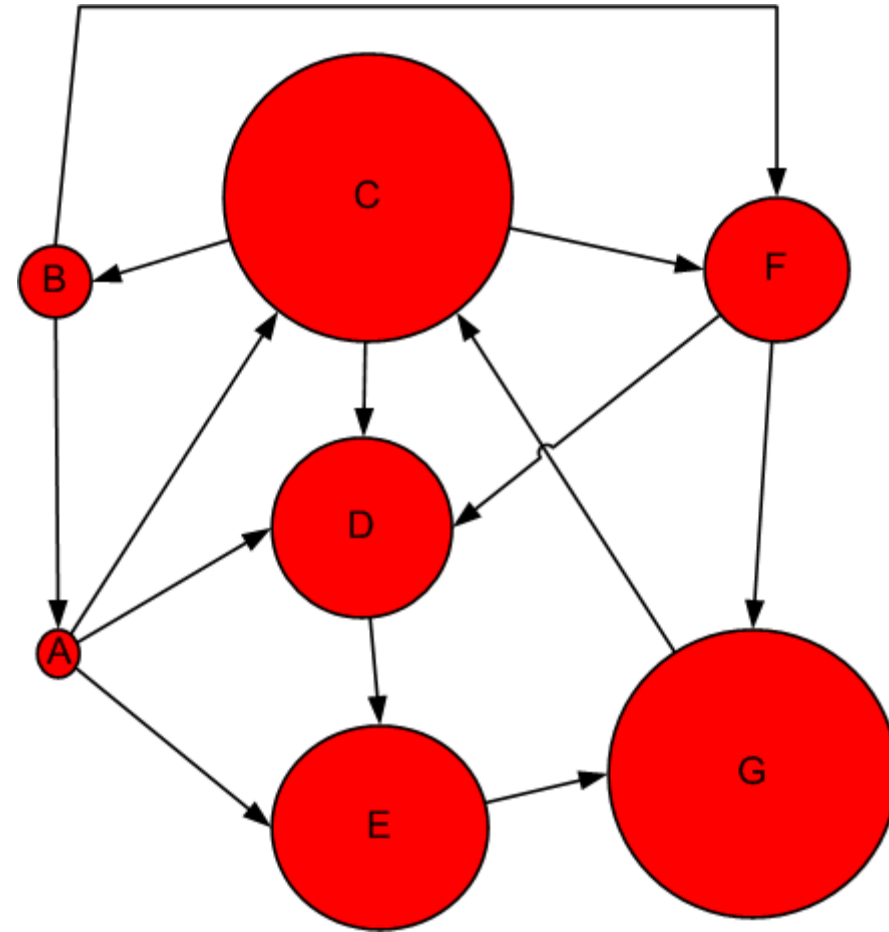
Step	A	B	C	D	E	F	G	Sum
1	0.143	0.143	0.143	0.143	0.143	0.143	0.143	1.000
2	0.071	0.048	0.190	0.167	0.190	0.119	0.214	1.000
3	0.024	0.063	0.238	0.147	0.190	0.087	0.250	1.000
4	0.032	0.079	0.258	0.131	0.155	0.111	0.234	1.000
5	0.040	0.086	0.245	0.152	0.142	0.126	0.210	1.000
6	0.043	0.082	0.224	0.158	0.165	0.125	0.204	1.000
7	0.041	0.075	0.219	0.151	0.172	0.115	0.228	1.000
8	0.037	0.073	0.241	0.144	0.165	0.110	0.230	1.000
9	0.036	0.080	0.242	0.148	0.157	0.117	0.220	1.000
10	0.040	0.081	0.232	0.151	0.160	0.121	0.215	1.000
11	0.040	0.077	0.228	0.151	0.165	0.118	0.220	1.000
12	0.039	0.076	0.234	0.148	0.165	0.115	0.223	1.000
13	0.038	0.078	0.236	0.148	0.161	0.116	0.222	1.000
14	0.039	0.079	0.235	0.149	0.161	0.118	0.219	1.000
15	0.039	0.078	0.232	0.150	0.162	0.118	0.220	1.000
Rank	7	6	1	4	3	5	2	

Effect of PageRank



PageRank

Node	Rank
A	7
B	6
C	1
D	4
E	3
F	5
G	2



**Centrality in terms of how you connect
others
(information broker)**

Betweenness Centrality

Another way of looking at centrality is by considering how important nodes are in connecting other nodes



Linton Freeman

$$C_b(v_i) = \sum_{s \neq t \neq v_i} \frac{\sigma_{st}(v_i)}{\sigma_{st}}$$

σ_{st} The number of shortest paths from vertex s to t – a.k.a. **information pathways**

$\sigma_{st}(v_i)$ The number of **shortest paths** from s to t that pass through v_i

Normalizing Betweenness Centrality

- In the best case, node v_i is on all shortest paths from s to t , hence,

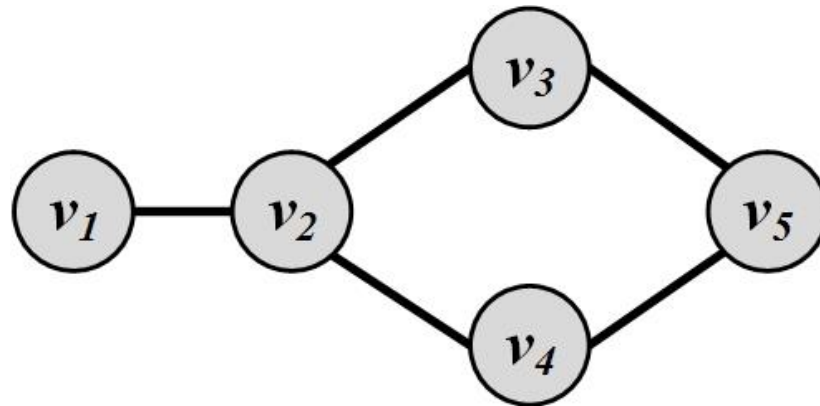
$$\frac{\sigma_{st}(v_i)}{\sigma_{st}} = 1$$

$$\begin{aligned} C_b(v_i) &= \sum_{s \neq t \neq v_i} \frac{\sigma_{st}(v_i)}{\sigma_{st}} \\ &= \sum_{s \neq t \neq v_i} 1 = 2 \binom{n-1}{2} = (n-1)(n-2) \end{aligned}$$

Therefore, the maximum value is $(n-1)(n-2)$

Betweenness centrality: $C_b^{\text{norm}}(v_i) = \frac{C_b(v_i)}{2 \binom{n-1}{2}}$

Betweenness Centrality: Example 1



$$C_b(v_2) = 2 \times \left(\underbrace{(1/1)}_{s=v_1, t=v_3} + \underbrace{(1/1)}_{s=v_1, t=v_4} + \underbrace{(2/2)}_{s=v_1, t=v_5} + \underbrace{(1/2)}_{s=v_3, t=v_4} + \underbrace{0}_{s=v_3, t=v_5} + \underbrace{0}_{s=v_4, t=v_5} \right) \\ = 2 \times 3.5 = 7,$$

$$C_b(v_3) = 2 \times \left(\underbrace{0}_{s=v_1, t=v_2} + \underbrace{0}_{s=v_1, t=v_4} + \underbrace{(1/2)}_{s=v_1, t=v_5} + \underbrace{0}_{s=v_2, t=v_4} + \underbrace{(1/2)}_{s=v_2, t=v_5} + \underbrace{0}_{s=v_4, t=v_5} \right) \\ = 2 \times 1.0 = 2,$$

$$C_b(v_4) = C_b(v_3) = 2 \times 1.0 = 2,$$

$$C_b(v_5) = 2 \times \left(\underbrace{0}_{s=v_1, t=v_2} + \underbrace{0}_{s=v_1, t=v_3} + \underbrace{0}_{s=v_1, t=v_4} + \underbrace{0}_{s=v_2, t=v_3} + \underbrace{0}_{s=v_2, t=v_4} + \underbrace{(1/2)}_{s=v_3, t=v_4} \right) \\ = 2 \times 0.5 = 1,$$

Computing Betweenness

- In betweenness centrality, we compute shortest paths between all pairs of nodes to compute the betweenness value.
- **Trivial Solution:**
 - Use Dijkstra and run it $O(n)$ times
 - We get an $O(n^3)$ solution
- **Better Solution:**
 - Brandes Algorithm:
 - $O(nm)$ for unweighted graphs
 - $O(nm + n^2 \log n)$ for weighted graphs

Closeness Centrality

- The intuition is that influential/central nodes can quickly reach other nodes
- These nodes should have a smaller average shortest path length to others

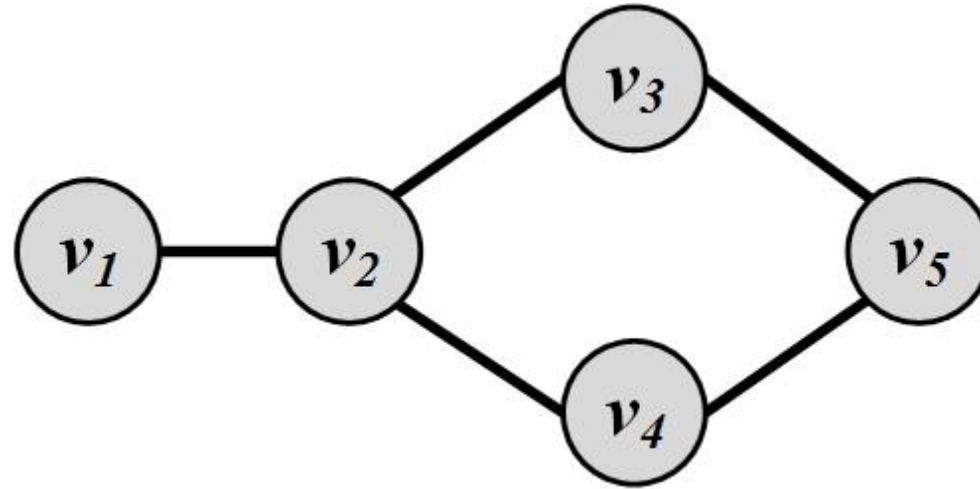


Linton Freeman

Closeness centrality: $C_c(v_i) = \frac{1}{\bar{l}_{v_i}}$

$$\bar{l}_{v_i} = \frac{1}{n-1} \sum_{v_j \neq v_i} l_{i,j}$$

Closeness Centrality: Example 1



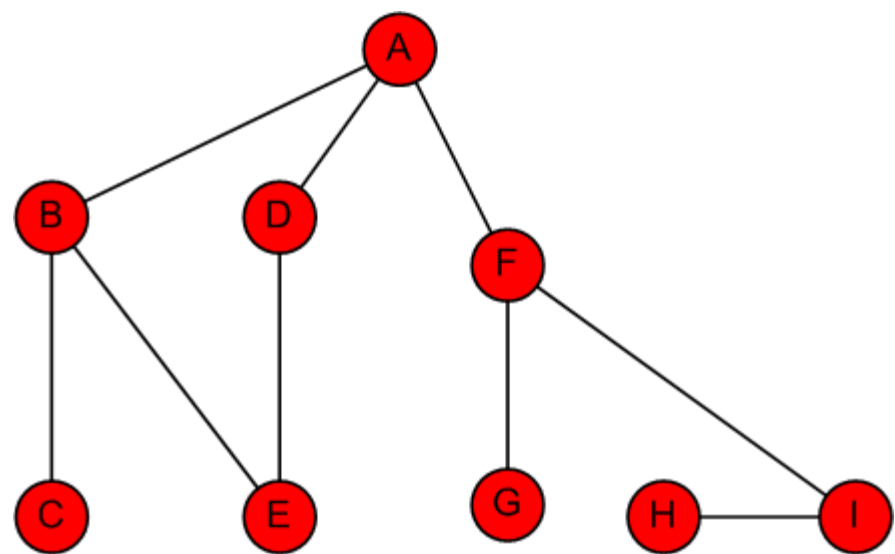
$$C_c(v_1) = 1 / ((1 + 2 + 2 + 3)/4) = 0.5,$$

$$C_c(v_2) = 1 / ((1 + 1 + 1 + 2)/4) = 0.8,$$

$$C_c(v_3) = C_b(v_4) = 1 / ((1 + 1 + 2 + 2)/4) = 0.66,$$

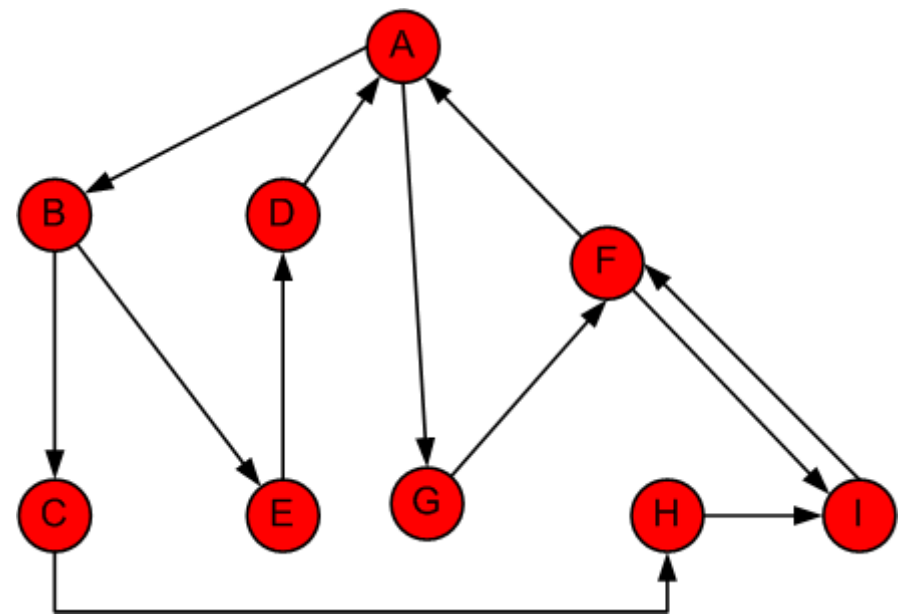
$$C_c(v_5) = 1 / ((1 + 1 + 2 + 3)/4) = 0.57.$$

Closeness Centrality: Example 2 (Undirected)



Node	A	B	C	D	E	F	G	H	I	D_Avg	Closeness Centrality	Rank
A	0	1	2	1	2	1	2	3	2	1.750	0.571	1
B	1	0	1	2	1	2	3	4	3	2.125	0.471	3
C	2	1	0	3	2	3	4	5	4	3.000	0.333	8
D	1	2	3	0	1	2	3	4	3	2.375	0.421	4
E	2	1	2	1	0	3	4	5	4	2.750	0.364	6
F	1	2	3	2	3	0	1	2	1	1.875	0.533	2
G	2	3	4	3	4	1	0	3	2	2.750	0.364	6
H	3	4	5	4	5	2	3	0	1	3.375	0.296	9
I	2	3	4	3	4	1	2	1	0	2.500	0.400	5

Closeness Centrality: Example 3 (Directed)



Nod	A	B	C	D	E	F	G	H	I	D Avg	Closeness Centrality	Rank
A	0	1	2	3	2	2	1	3	3	2.125	0.471	1
B	3	0	1	2	1	4	4	2	3	2.500	0.400	2
C	4	5	0	7	6	3	5	1	2	4.125	0.242	9
D	1	2	3	0	3	3	2	4	5	2.875	0.348	3
E	2	3	4	1	0	4	3	5	5	3.375	0.296	6
F	1	2	3	4	3	0	2	4	4	2.875	0.348	3
G	2	3	4	5	4	1	0	5	2	3.250	0.308	5
H	4	4	5	6	5	2	4	0	1	3.875	0.258	8
I	2	3	4	5	4	1	4	5	0	3.500	0.286	7

An Interesting Comparison!

Comparing three centrality values

- Generally, the 3 centrality types will be positively correlated
- When they are not (or low correlation), it usually reveals interesting information

	Low Degree	Low Closeness	Low Betweenness
High Degree		<i>Node is embedded in a community that is far from the rest of the network</i>	<i>Ego's connections are redundant - communication bypasses the node</i>
High Closeness	<i>Key node connected to important/active alters</i>		<i>Probably multiple paths in the network, ego is near many people, but so are many others</i>
High Betweenness	<i>Ego's few ties are crucial for network flow</i>	<i>Very rare! Ego monopolizes the ties from a small number of people to many others.</i>	

This slide is modified from a slide developed by James Moody

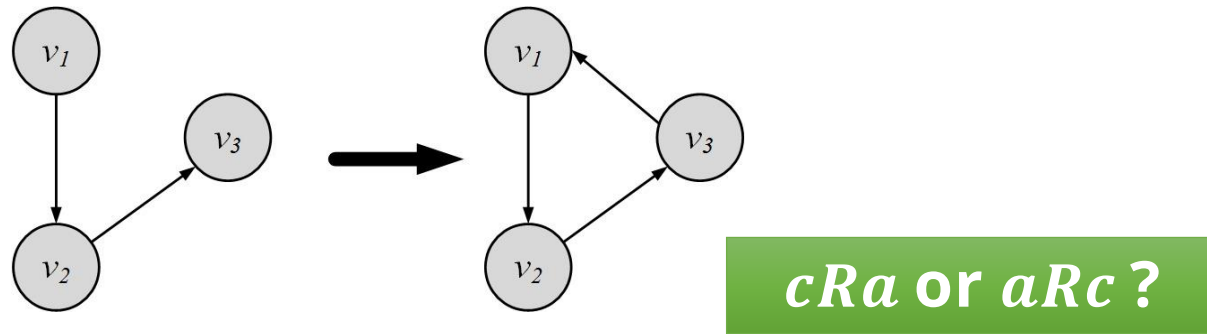
Friendship Patterns

- **Transitivity/Reciprocity**
- **Status/Balance**

I. Transitivity and Reciprocity

Transitivity

- Mathematic representation:
 - For a transitive relation $R: aRb \wedge bRc \rightarrow aRc$

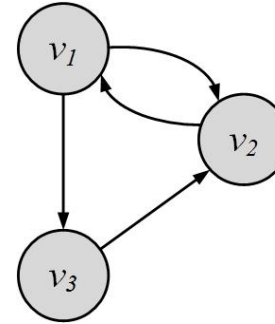


- In a social network:
 - ***Transitivity is when a friend of my friend is my friend***
 - Transitivity in a social network leads to a denser graph, which in turn is closer to a complete graph
 - We can determine how close graphs are to the complete graph by measuring transitivity

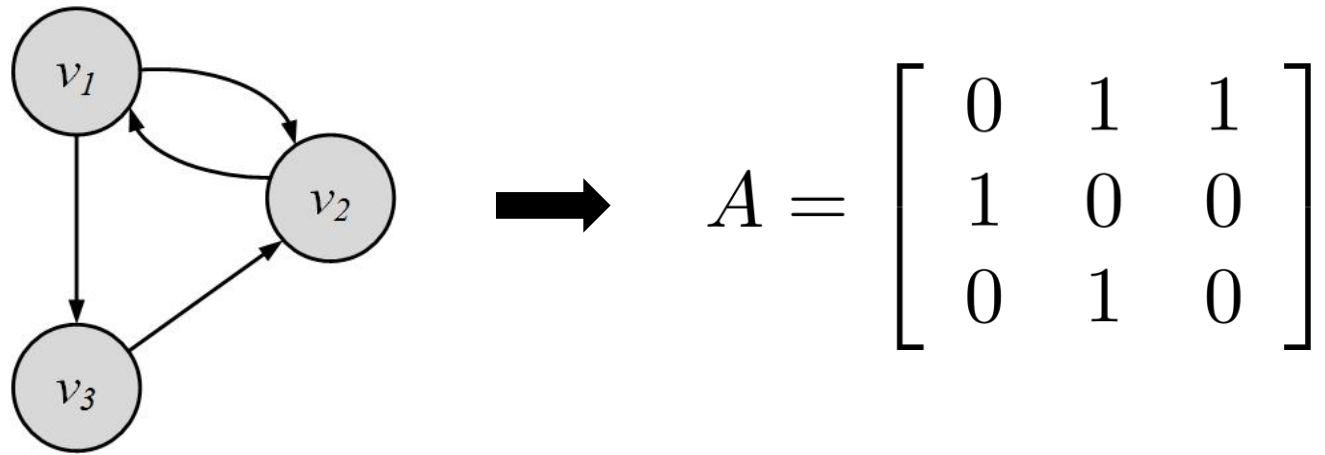
Reciprocity

***If you become my friend,
I'll be yours***

- Reciprocity is simplified version of transitivity
 - It considers closed loops of length 2
- If node v is connected to node u ,
 - u by connecting to v , exhibits reciprocity



Reciprocity: Example



$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

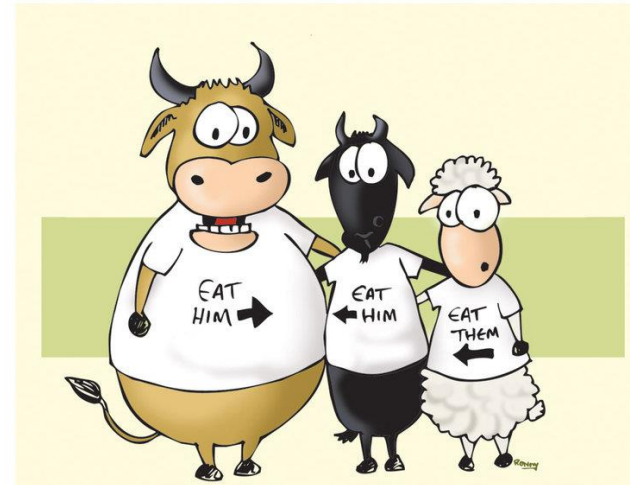


Reciprocal nodes: v_1, v_2

$$R = \frac{1}{m} \text{Tr}(A^2) = \frac{1}{4} \text{Tr} \left(\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \right) = \frac{2}{4} = \frac{1}{2}.$$

II. Balance and Status

- Measuring consistency in friendships



Social Balance Theory

Social balance theory

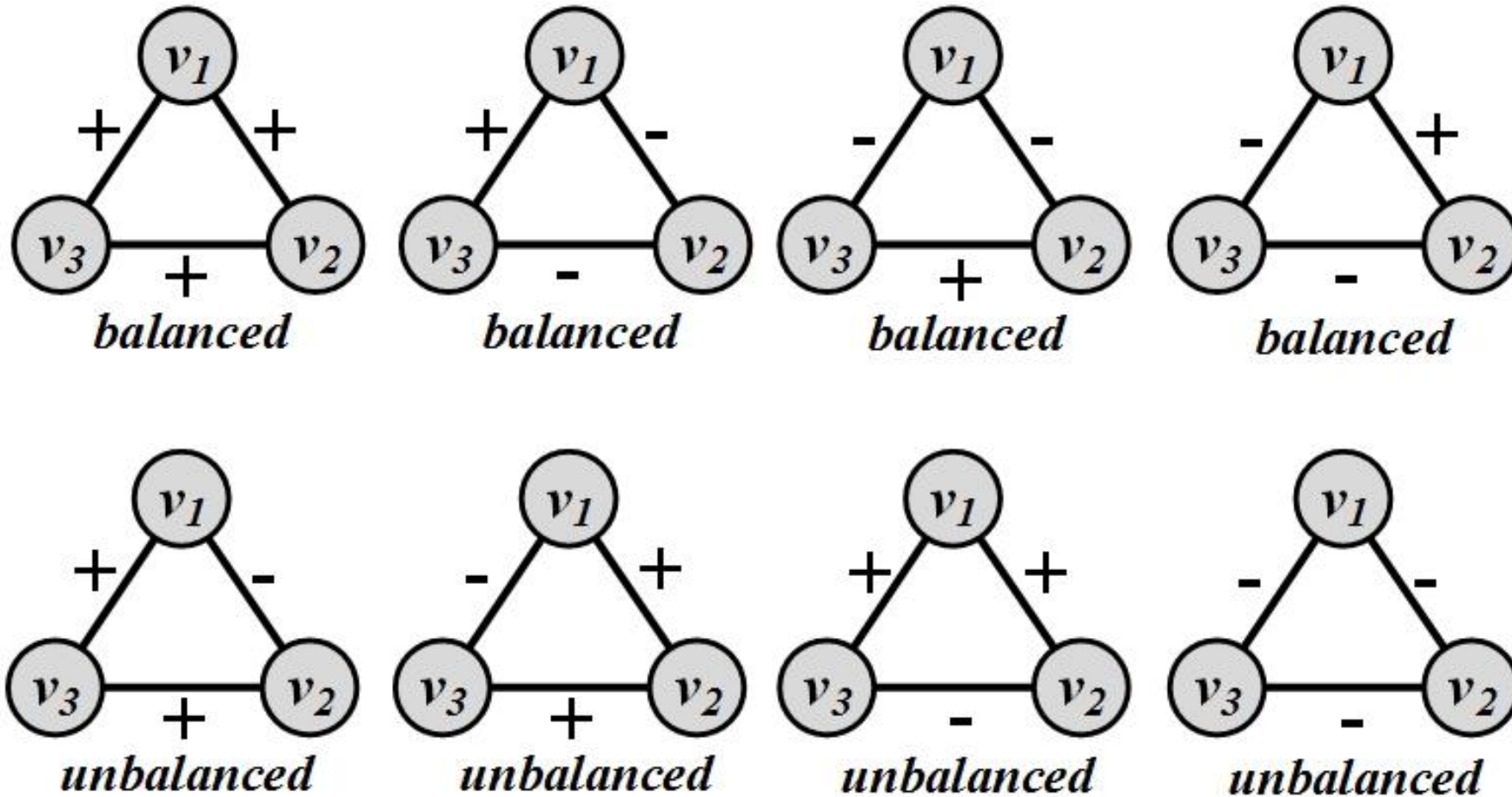
- Consistency in friend/foe relationships among individuals
- Informally, friend/foe relationships are consistent when

*The friend of my friend is my friend,
The friend of my enemy is my enemy,
The enemy of my enemy is my friend,
The enemy of my friend is my enemy.*

- In the network
 - Positive edges demonstrate friendships ($w_{ij} = 1$)
 - Negative edges demonstrate being enemies ($w_{ij} = -1$)
- Triangle of nodes i, j , and k , is balanced, if and only if
 - w_{ij} denotes the value of the edge between nodes i and j

$$w_{ij}w_{jk}w_{ki} \geq 0.$$

Social Balance Theory: Possible Combinations



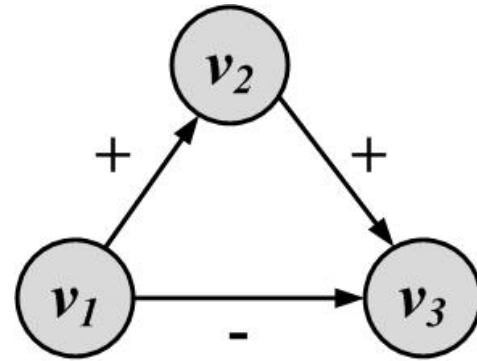
For any cycle, if the multiplication of edge values become positive, then the cycle is socially balanced

Social Status Theory

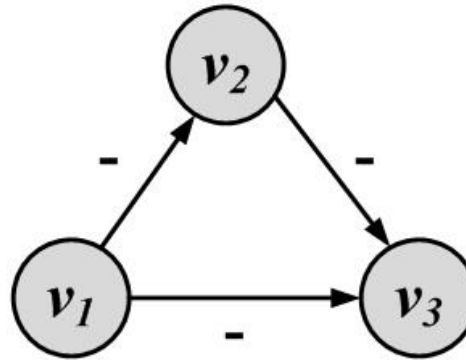
- **Status:** how prestigious an individual is ranked within a society
- **Social status theory:**
 - How consistent individuals are in assigning status to their neighbors
 - Informally,

*If X has a higher status than Y and Y has a higher status than Z,
then X should have a higher status than Z.*

Social Status Theory: Example



Unstable configuration



Stable configuration

- A directed '+' edge from node X to node Y shows that Y has a higher status than X and a '-' one shows vice versa

Similarity

How similar are two nodes in a network?

- **Structural Equivalence**
- **Regular Equivalence**

Structural Equivalence

- **Structural Equivalence:**

- We look at the neighborhood shared by two nodes;
- The size of this shared neighborhood defines how similar two nodes are.

- ***Example:***

- *Two brothers have in common*
 - *sisters, mother, father, grandparents, etc.*
- *This shows that they are similar,*
- *Two **random** male or female individuals do not have much in common and are dissimilar.*

Structural Equivalence: Definitions

- **Vertex similarity:** $\sigma(v_i, v_j) = |N(v_i) \cap N(v_j)|$

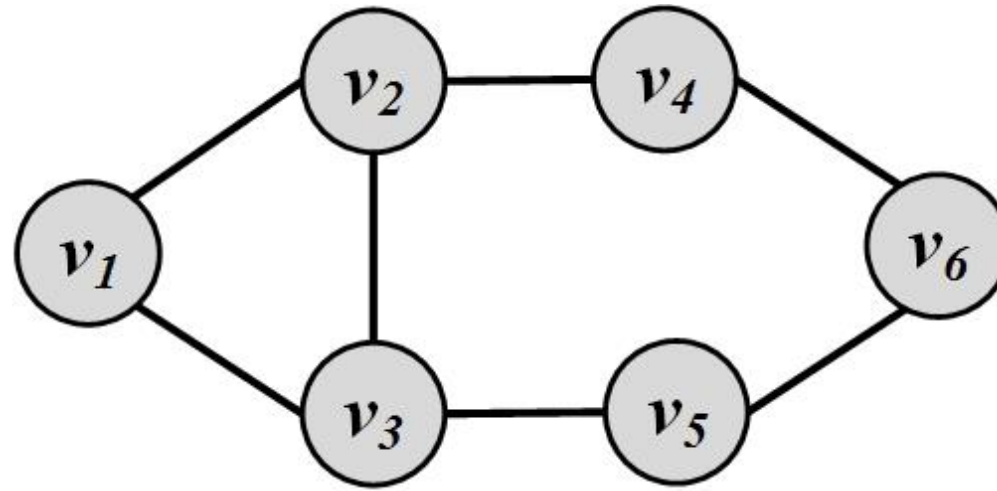
Normalize?

Jaccard Similarity: $\sigma_{Jaccard}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|}$

Cosine Similarity: $\sigma_{Cosine}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{\sqrt{|N(v_i)| |N(v_j)|}}$

- The neighborhood $N(v)$ often excludes the node itself v .
 - **What can go wrong?**
 - Connected nodes not sharing a neighbor will be assigned zero similarity
 - **Solution:**
 - We can assume nodes are included in their neighborhoods

Similarity: Example



$$\sigma_{\text{Jaccard}}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{|\{v_1, v_3, v_4, v_6\}|} = 0.25$$

$$\sigma_{\text{Cosine}}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{\sqrt{|\{v_1, v_3, v_4\}| |\{v_3, v_6\}|}} = 0.40.$$

Measuring Similarity Significance: compare the calculated similarity value with its expected value where vertices pick their neighbors at random

- For vertices v_i and v_j with degrees d_i and d_j this expectation is $d_i d_j / n$
 - There is a d_i / n chance of becoming v_i 's neighbor
 - v_j selects d_j neighbors
- We can rewrite neighborhood overlap as

$$\sigma(v_i, v_j) = |N(v_i) \cap N(v_j)| = \sum_k A_{i,k} A_{j,k}$$